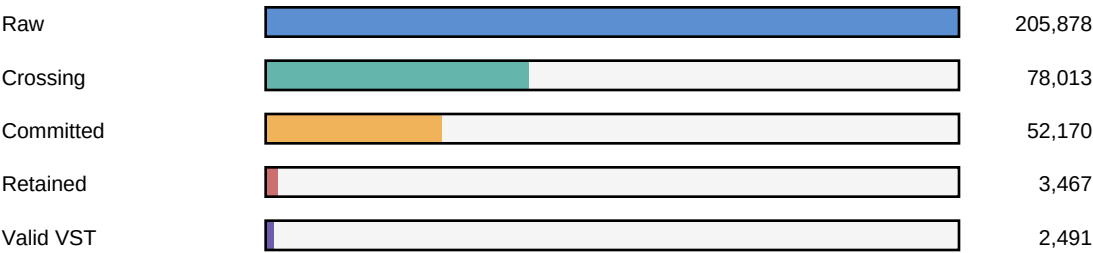


VSTF HMM Benchmark: activity is not outcome

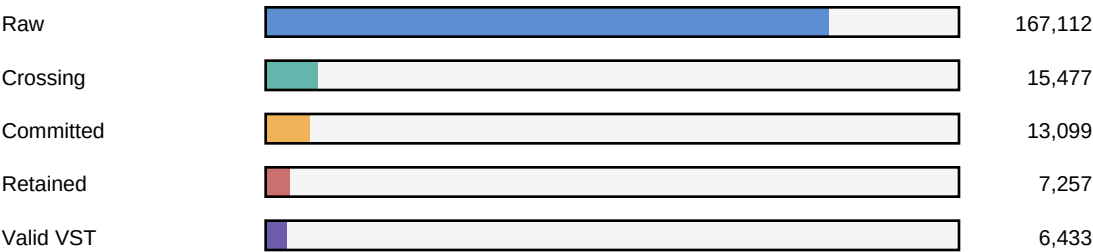
10,000 trajectories per system; seed=20260917; validation lag=2; retention window=6

Nested event attrition

System A: high activity, fragile retention



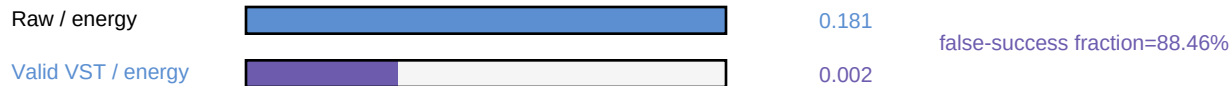
System B: moderate activity, retained outcomes



Efficiency ranking

Raw activity, crossings and valid VST per energy are normalized to the best system.

System A: high activity, fragile retention



System B: moderate activity, retained outcomes



Interpretation

System A produces more raw activity and crossings, but many events fail validation or retention and its cost boundary is higher. System B ranks lower on raw activity, but higher on retained valid VST yield per supplied energy. This is the incremental VSTF claim in miniature: the acceptance layer changes the scientific ranking.